|INCH-POUND|

A-A-58050
20 October 1995
SUPERSEDING
MIL-F-38762(USAF)
26 April 1966

COMMERCIAL ITEM DESCRIPTION

FLUORESCENT PENETRANT INSPECTION UNIT
TYPE MA-2

The General Services Administration has authorized the use of this commercial item description as a replacement for Type MA-2 of MIL-F-38762(USAF) for all federal agencies

1 SCOPE This commercial item description covers the MA-2 type of fluorescent penetrant inspection unit The type MA-2 unit has the following internal tank dimensions: 29 inches wide, 43 inches long, and 26 inches deep. These units will be used to inspect aircraft components with a maximum weight of 300 pounds using the post-emulsification hydrophilic fluorescent penetrant method

2 SAI IFNT CHARACTFRISTICS

2 1 First Article Testing This commercial item description makes provisions for first article testing

2.2 Design and Construction. The units shall be designed and constructed to facilitate efficient inspection of parts and assemblies capable of being penetrant inspected They shall be constructed so as to hold aircraft parts and inspection medium without deformation of the components The units shall be designed and constructed so that during the normal inspection process there will be no leakage of penetrant, remover, developer, and/or water within the area bounded by the outermost extremities of the inspection unit Possible points where leakage might occur are Between the penetrant, remover, and developer stations and their respective drain stations, between the backsplash and tank frame on the rinse station, runoff from station covers while in the open position

2 3 Tolerances All tolerances for dimensional specifications listed in this purchase description are ±1/2 inch unless otherwise noted

2.4 Stations. This inspection unit shall consist of the following stations: penetrant, penetrant drain, pre-rinse, remover, post drip and hand touchup, developer, developer drain, dryer, rest and inspection stations

2 4 1 Penetrant. Remover and Developer Stations These stations shall consist of a frame, tank, cover, and slatted wooden grille A 1 1/4 inch drain pipe connection and cutoff valve shall be provided at the bottom of the tanks

2 4.1 1 Station Dimensions The usable area inside the tank shall be at least 29 inches wide, 43 inches long, and 26 inches deep. The frame shall have overall outside dimensions not to exceed 48 inches in length, 34 inches in width, and 36 inches in height (excluding cover and cover handle) All tanks shall be fabricated of 16 gauge stainless steel

Beneficial comments(recommendations, additions, deletions) and any pertinent data which may be of use in improving the document should be addressed to San Antonio ALC/TILBD, Bldg 171 Kelly AFB TX 78241-6425 Document Improvement Proposal(DD Form 1426) appearing at the end of this document or by letter

FSC 6635

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2 4 1 2 Remover Station The remover station shall consist of a frame, tank, and air agitation system The agitation system shall include an air distribution manifold, manifold shutoff valve, air pressure regulator, and an air pressure gauge All controls and gauges(i e manifold shutoff valve, air pressure regulator and pressure gauge) shall be readable from the front of the remover station The agitation shall allow thorough agitation of the remover solution throughout the tank and shall not cause the remover solution to foam

2 4 2 Pre-rinse Station The pre-rinse shall consist of a frame tank, backsplash, slatted wooden grille, and a hand hose with nozzle The station shall be plumbed to provide both hot and cold water to the hand hose Separate controls for water temperature, pressure regulation, and flow regulation shall be provided and shall be located on the front of the station Gauges to measure water temperature and pressure shall be located on or at the front of the station The hand hose shall be long enough to reach into the each tank when the tanks are butted side by side A 1 1/4 inch drain pipe connection and cutoff valve shall be provided at the bottom of the tanks

2 4 2 1 Backsplashes The backsplashes shall be provided to prevent overspray from the rinse station from reaching the rear of the unit It shall be fabricated from corrosion resistant material

2 4 3 Post Drip and Hand Touch Up Station The post drip and hand touchup rinse station shall consist of frame, tank, backsplash, slated wooden grille, blacklight, full canopy to exclude light and a hand hose with nozzle The station shall be plumbed to provide both hot and cold water to the hand hose Separate controls for water temperature, pressure regulation and flow regulation shall be provided and shall be located on the front of the station Gauges to measure water temperature and pressure shall be located on the front of the station The blacklight shall be mounted on the backsplash of the station The canopy frame shall be constructed such that a technician can hand carry small parts unobstructed from the remover station to the post drip station and from the post drip station to the wet developer station i e there shall be no canopy support post in the front of the station A 1 1/4 inch drain pipe connection and cutoff valve shall be provided at the bottom of the tanks

2 4 3 1 Canopy. The post drip and hand touchup station requires a full canopy with side curtains, split to allow ingress and egress through the station The curtain material must be UL or equivalent certified tear and fire resistant to 1000°F The canopy must provide a split top to allow access from an overhead conveyor The canopy shall allow not more than 20 lux of light from the outside

2 4 4 Penetrant and Developer Drain Stations All drain stations shall consist of a frame, drain pan, and a full roller top grille The drain stations shall be structurally supported from below Each station will conform to the requirements in paragraph 4.3.6. Design shall be such to insure that the penetrant or developer is completely directed back into its tank. The drain pan shall prevent fluids from running off inspected materials from dripping on the floor The length and width of the roller top grille shall conform to the length and width of the station and shall conform to the height of adjacent stations

2 4 5 Dryer Station The dryer station shall incorporate a full roller grille to facilitate the movement of parts from the developer drain station to the rest station The dryer cabinet shall be designed with a split roof to allow access from an overhead conveyor and with both ends opened Split curtains shall be provided for each end

2 4 5 1 Dryer Dimensions. The interior area above the grille shall be at least 30 inches wide, 72 inches long, and 25 inches high The curtain opening shall be at least 30 inches by 25 inches The exterior dimensions shall be 34 inches wide and 72 long

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2 4 5.2 Type MA-2 Heater Capacity The heater shall be at least 8000 watt element

2 4 5 3 Thermostats The thermometer-thermostat shall be scaled from 50°F to 250°F in increments of 10°F The thermostat shall control the dryer temperature to within plus or minus 10°F of the setting

2 4 6 Rest Station The station shall be similar to the drain stations (paragraph 2 4 4) except the drain pan shall be omitted

2.4 7 Inspection Station The inspection station shall provided with a full roller grille table top, dust pan, exhaust fan, two black lights, a white light, and a canopy. The canopy shall allow access of an overhead conveyor.

2 4 7.1 The dust pan shall be located directly under the table top for the exclusion of light It shall cover the entire area below the table top

2 4 7 2 Roller top grille, The roller grille table top shall have a material stop that can be positioned at either end of the inspection station The full roller grille table top working width and length shall be at least 31 inches and 72 inches, respectively The table top base frame width shall be 36 inches The overall dimensions of the inspection station shall be 72 inches long, 60 inches wide, and at least 76 inches high

2.5 Voltage. The dryer heater elements shall be capable of operating from both 230V and 460V, 50 and 60 Hz, single phase AC source The initial line operating voltage shall be as specified by the procuring activity The rest of the equipment shall operate from 110-120V, 50 and 60 Hz, single phase AC source The unit shall require a single connection to each source (one 120V and one 230V or 460V)

2 6 Controls Each black light, white light, exhaust fan, and the dryer heater circuits shall operate independently of any other unit and shall have a separate control or switch Each line control and heater or dryer control shall be equipped with an indicator light

2 7 Nozzles. The nozzles shall have a spring-operated, lever-type hand valve. Nozzles in the pre-rinse and hand touchup stations shall be of the fan-shaped course spray type Nozzles that have adjustable spray patterns shall not be used. The delivery rate of the nozzles shall be not less than 3 gallons per minute and not more than 6 gallons per minute at 40 PSI pressure.

2 8 Black Lights. The black lights which are employed to fluoresce the inspection medium shall have an intensity of 1000$\mu W/cm^2$ at a distance of 15 inches. The black light wavelength shall be in the range of 320 to 380 nm with a peak intensity at 365 nm The lights shall be supported by adjustable brackets and directed toward the part to be inspected The lights shall be readily dismountable from the adjustable brackets without removing or loosening any fasteners or fixtures.

2 8 Baskets. Six wire baskets, with handles, made of corrosion resistant material shall be provided Unless otherwise specified, the baskets shall have an internal volume of 452 cubic inches and maximum mesh opening of 3/4 inches

3 Regulatory requirements The contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable

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3 1 Metric products Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pounds unit, provided they fall within specified tolerances using conversion tables contained in the
latest revision of Federal Standard No 376, and all other requirements of this Commercial Item Description are met If a product is manufactured to metric dimensions and those dimensions exceed tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable

4 QUALITY ASSURANCE PROVISIONS

4 1 Contractor Certification The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producer's own drawings, standards, and quality assurance practices The government reserves the right to require a government witnessed, pre-award, sample verification test at the contractor's site to insure conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract

4 2 First article inspection, First article inspection shall be performed on one complete fluorescent penetrant unit when a first article sample is required (see 3 1 and 7 2) This inspection shall include the requirements of paragraph 5 3 Test Methods

4 2 1 Acceptance Tests Acceptance tests are intended to verify that production units are similar to the applicable specifications The following are the minimum inspection requirements for production acceptance· paragraph 4 3 1 and paragraph 4.3.2

4 3 Test Methods The following test shall be conducted in an ambient temperature range of 68-95 degrees F and in the order given

4 3 1 Examination of Product The inspection units selected for inspection shall be inspected during manufacture and upon completion to determine compliance with the requirements of the contract and this specification with respect to material, workmanship, and dimensions

4 3 2 Tanks. All tanks, drain pans, associated lines, outlet valves, etc , shall be penetrant inspected with materials listed on MIL-I-25135 Each tank and drain pan shall have one of the penetrants of Type I, Level 2, Method D, applied to the interior seams, bends, and connections The penetrant shall dwell for at least 30 minutes The developer, Form A, C, or D shall be applied to the exterior of these seams, bends, and connections All exterior seams, bends, and connections shall be inspected under black light for indication of penetrant leakage This inspection shall be accomplished under low light (ambient light < 20lux) with a blacklight conforming to the characteristics of paragraph 3 10 Any leakage shall be cause for rejection

4 3 2 1 Drain Pans The drain pan outlet will be plugged and the pan filled with water The outlet will be unplugged and allowed to drain into its associated inspection tank Incomplete drainage shall be cause for rejection

4 3 3 Dryer Stations

4 3 3 1 Temperature rise The thermostat controlling the temperature within the dryer cabinet shall be set at 250°F The interior of the dryer cabinet shall reach a temperature of 225°F from an ambient room temperature of 70°F in a maximum of 40 minutes The temperature shall be checked in five locations within the dryer cabinet, using an ASTM-E1 type thermometer or equivalent

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4.3.3.2 Thermostats After the test for temperature rise has been completed, the thermostat shall be set at 225°F and the dryer cabinet stabilized at this temperature for one hour The thermometer specified in 4.3.3.1 shall then be used to check the temperature at the thermostat element, at each end within 2 inches of the curtains, and at the center of the upper shelf four inches from the near edge (a total of four readings) The thermometer shall remain at each location for ten minutes prior to reading the temperature The average of the four readings shall be within ±10% of the thermostat setting An average outside of the range 225°F ± 10% shall be cause for rejection

4 3 3 3 Temperature recovery the test for temperature recovery shall be conducted while the temperature of the dryer cabinet is still stabilized The curtains of the cabinet shall be opened to maximum for one minute and then closed The temperature shall be measured in the locations stated in 4 3 3.2 An average temperature outside the range 225 degrees F ±10% shall be cause for rejection

4 3 4 Liquid environmental test A coarse water spray shall be directed to all electrical components including switches and wiring enclosure, located on the drain, penetrant, remover, rinse, developer, and dryer stations from all possible angles of attack All electrical equipment shall be operated during this test Failure of any equipment to operate satisfactorily or the accumulation of any water within the enclosure protecting the electrical components including switches and wiring shall be cause for rejection The government inspector shall designate the locations to be inspected for water accumulation

4 3 5 Delivery rate of liquid flow nozzles. The delivery rate of the liquid in the pre-rinse and post rinse stations shall be determined by measuring the amount of water passing through the nozzle for a period of one minute The nozzles shall be opened to maximum The delivery rate of the liquid in the rinse station shall be determined by measuring the amount of water at 40 PSI passing through the spray nozzle in one minute A coarse spray type with a flow rate of less than 3 gallons per minute or greater than 6 gallons per minute shall be cause for rejection

4 3 6 Roller Top Grilles

4 3 6 1 Load test Each roller top grille shall be load tested with the grille assembly supported in its usual manner Place a load of at least 300 pounds at one end of the grille uniformly distributed over four rollers (two axles) All rollers shall roll freely under this load traveling at a rate not to exceed six inches per second Evidence of failure or damage of the grille assembly or rollers not rolling freely under load shall be cause for rejection.

4 3 6.2 Roll test The following roll test shall be performed after the load test and on each roller top grille The grille shall be supported so that the top line of the rollers is at an angle not to exceed seven degrees with respect to the horizontal. A 50 pound load, uniformly distributed over four rollers (two axles) on the raised end of the grille, should move freely and smoothly to the lower end with an initial external force to start it Failure of the load to move freely and smoothly to the lower end shall be cause for rejection

4 3 7 Black light test The test for light intensity shall be measured with an ultraviolet intensity meter and shall measure at least 1000 microwatts per square centimeter at a distance of 15 inches from the light with a peak amplitude at 365 manometers Intensities less than 1000 uw/cm^2 shall be cause for rejection

5 Preservation packaging. packing. labeling. and marking Preservation, packaging, packing, labeling, and marking shall be specified in the contract or order

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6 0 NOTES

6 1 Part Numbering Identification The following part identification numbering procedure is the government purposes and does not constitute requirement for the contractor

a MA-2

6 2 Source of Documents

6.2 1 MIL-I-25135
Inspection Materials, Penetrants
Defense Printing Services Detachment Office, Order Desk
700 Robbins Ave
Philadelphia, PA 19111-5094

6.2 2 ASTM-E1, ,
Standard Specification for ASTM Thermometers
American Society for Testing and Materials
100 Barr Harbor Drive
West Conshokocken Pa 19428

6 3 Ordering data

a Title, number, and date of this description.
b Applicable CID based part number (see 6 1)
c Specify special packaging requirements
d Contracting officer may request proof of certification of commercial item prior to first contract delivery
e Requirement for first article testing See 2 1 and 4 2)

6 3 National Stock Number

a MA-2 NSN 6635-00-110-9463

6 4 Known sources of supply

MAGNAFLUX
Division of Illinois Tool Works Inc
7301 W Ainsile Ave
Harwood Heights IL 60656

Gould-Bass
1431 W Second St
Pomana CA 91766

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MILITARY INTEREST

CUSTODIANS
Air Force-99
Navy-AS
Army-GL
DLA-GS

CIVIL AGENCY COORDINATING ACTIVITY
GSA-FSS

PREPARING ACTIVITY
AIR FORCE-82

AGENT
AIR FORCE-99

PROJECT NUMBER 6635-0163

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I RECOMMEND A CHANGE:

1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
A-A-58050	951026

3. DOCUMENT TITLE: FLUORSCENT PENTRANT INSPECTION UNIT TYPE MA2

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)
AE SA-ALC/TILDD	(1) Commercial (512) 925-6314 (2) AUTOVON 945-6314
c. ADDRESS (Include Zip Code) BLDG 207 KELLY AFB TX 78241-6425	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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